THIRU KUMARAN T
M.Pharm – Pharmaceutical Biotechnology
+91-9944735972
Gmail
LinkedIn | GitHub | Portfolio



PROFESSIONAL SUMMARY:

M.Pharm graduate in Pharmaceutical Biotechnology with hands-on experience in pharmaceutical R&D and Analytical Development, complemented by practical exposure to bioinformatics and computational biology. Skilled in biological data retrieval, sequence analysis, and genomic data analysis using Linux command-line tools, biological databases. Possess practical knowledge of genomics, transcriptomics, microbial genomics, and biological data formats, along with capabilities in experimental data interpretation, and analytical assay evaluation. The combination of pharmaceutical biotechnology and computational biology provides me a multidisciplinary foundation for contributing to genomic data analysis, biological research, and the interpretation of computational findings in the context of experimental and pharmaceutical applications.

PROFESSIONAL EXPERIENCE:

Biological E Limited, Hyderabad October 2025 – March 2026

Trainee – R&D, Analytical Development

- Performed analytical characterization of vaccine-related Poliovirus-Like Particle (PVLP) samples using HCP, **HCD, BCA, D-Antigen ELISA, and Serum Neutralization assays**.
- Generated, recorded, organized, and analyzed experimental data from multiple analytical assays to support characterization and evaluation of vaccine-related samples.
- Interpreted analytical results against predefined acceptance criteria and evaluated assay performance and experimental outcomes.
- Evaluated analytical parameters including **linearity, precision, and accuracy** based on experimental assay data. Maintained experimental records and prepared analytical reports to support documentation, review, and R&D activities.

M.PHARM DISSERTATION:

Biopharmaceutical Characterization and Functional Potency Assessment of Poliovirus-Like Particles (PVLP)

- Conducted biopharmaceutical characterization of Poliovirus-Like Particle (PVLP) vaccine samples to evaluate product quality and functional potency.
- Utilized analytical assays including Host Cell Protein (HCP), Host Cell DNA (HCD), BCA protein assay, D-Antigen ELISA, and Serum Neutralization Assay.
- Generated, organized, and interpreted experimental data to assess vaccine sample characteristics and functional activity.
- Evaluated analytical method performance through linearity, precision, and accuracy assessments. Interpreted analytical results against predefined acceptance criteria to support product quality evaluation.

RESEARCH PROJECTS:

scRNA-seq Data Analysis of Human Glioblastoma - Research blog | GitHub

- Analyzed a human glioblastoma single-cell RNA sequencing (scRNA-seq) dataset using R and Seurat to characterize cellular populations.
- Performed quality control, normalization, highly variable gene identification, PCA, clustering, UMAP visualization, and marker-gene analysis.
- Identified and characterized 12 cell populations from 2,847 cells based on marker-gene expression.
- Developed a reproducible command-line workflow for scRNA-seq analysis.

APOE SNP Analysis and Association Study:

- Curated APOE variants from NCBI dbSNP, 1000 Genomes, and ADSP datasets to investigate genetic variation.
- Performed population-level genetic analysis using PLINK, including PCA and allele-frequency analysis.
- Evaluated selected variants using SIFT to investigate potential functional effects.
- Explored potential structural implications using SWISS-MODEL.

Genomic Investigation of Antibiotic Resistance in Klebsiella pneumoniae:

Whole Genome sequencing | Microbial Genomics

- Retrieved publicly available genomic sequencing data and assessed associated sample metadata.
- Performed raw sequencing data quality assessment using FastQC and interpreted sequencing quality metrics.
- Explored genome assembly, assembly quality assessment, and genome annotation workflows.
- Investigated antimicrobial resistance genes and their potential association with antibiotic resistance.

QC & Batch Release Analytics - GitHub

- Generated a synthetic dataset of 50 vaccine/biologics batches using Python, incorporating assay-specific measurement ranges and predefined acceptance criteria for HCP, HCD, BCA and D-Antigen.
- Applied statistical distributions and controlled variations to simulate realistic assay measurements and out-of-specification results.
- Performed data cleaning, exploratory data analysis and statistical evaluation using Python (Pandas, NumPy).
- Identified out-of-specification results, classified batches for quality review and evaluated assay-wise data distributions and batch quality trends.

ACADEMIC CREDENTIALS:

Master of Pharmacy (M. Pharm) – Pharmaceutical Biotechnology
JSS College of Pharmacy, Ooty
2024 - 2026 | CGPA: 9.42/10

Bachelor of Pharmacy (B. Pharm)
Arulmigu Kalasalingam College of Pharmacy, Krishnankovil
2018 – 2022 | CGPA: 7.14/10

TECHNICAL SKILLS:

Bioinformatics:
WGS, NGS data analysis, sequence analysis, genomics, transcriptomics, microbial genomics, variant analysis and biological data interpretation.

Programming & Computational Tools:
Python, R, SQL, Bash, Linux command line, Git, GitHub and Docker.

Wet Lab & Experimental Techniques:
HCP, HCD, BCA protein assay, D-Antigen ELISA, Serum Neutralization Assay, MIC, MBC, microbial culture techniques, aseptic techniques and microbiological analysis.

TRAINING & CERTIFICATIONS:

- **Hands-On Training on Cell Culture Techniques** — International Research Centre, Sathyabama University, January 03–13, 2025.
- **NPTEL Online Course and Examination on Intellectual Property** — Successfully completed.
- **E-Poster Presentation:** *"Molecular Characterization of APOE Gene SNPs in Alzheimer's Disease using Comparative Genomics and Population-Based Analysis"* — JSS College of Pharmacy, Ooty.
- **Organizing Committee Member**, International Conference **"PHARMANCIA"** — JSS College of Pharmacy, Ooty, March 7–8, 2025.
- **Two-Day Offline Workshop on Stem Cell Research** — EduTech Life, in association with Mechanica 2025, IIT Madras.

DECLARATION:

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.